Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 110601

Subject: Measurements and Instrumentation

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) The smallest change in a measured variable to which the instrument responds is
- (i) Sensitivity
- (ii) Drift
- (iii) Resolution
- (iv) Dead-band

(b) Which type of error cannot be eliminated?
- (i) Gross
- (ii) Random
- (iii) Systematic
- (iv) Zero error

(c) Higher C_p value means
- (i) Better process capability
- (ii) High variation
- (iii) Poor capability
- (iv) Low precision

(d) Hall sensor is used for measuring
- (i) Temperature
- (ii) Pressure
- (iii) Magnetic field
- (iv) Flow

(e) Hot wire anemometer is used for
- (i) Air velocity
- (ii) Pressure
- (iii) Displacement
- (iv) Torque

(f) Loading effect is lower in
- (i) High resistance voltmeter
- (ii) High resistance ammeter
- (iii) Low resistance voltmeter
- (iv) None

(g) Moving iron instrument is used for
- (i) AC only
- (ii) Both AC and DC
- (iii) DC only
- (iv) None

(h) LCR meter is used to measures
- (i) Voltage
- (ii) Inductance & Capacitance
- (iii) Resistance
- (iv) All

(i) Which instrument measures peak voltage?
- (i) DMM
- (ii) Megger
- (iii) CRO
- (iv) Clamp meter

(j) Sonography is based on
- (i) Ultrasound
- (ii) Magnetic resonance
- (iii) X-rays
- (iv) CT techniques

Q.2 (a) Differentiate between true value, measured value, and error. How do these affect the reliability of instruments? [7]

(b) What is resolution? Explain how resolution affects digital and analog measurement systems. [7]

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Q.3 (a) Discuss static sensitivity, scale readability, and drift with examples from temperature sensors. [7]

(b) What is Six-Sigma? Explain its significance in measurement system analysis. [7]

Q.4 (a) Explain the calculation of C_p and C_{pk} for a manufacturing process with an illustrative example. [7]

(b) Explain the working principle of RTD, and thermistor for temperature measurement. Compare their characteristics. [7]

Q.5 (a) Describe the construction and working of strain-gauge-based pressure transducers. [7]

(b) Explain the difference between active and passive transducers with suitable examples. [7]

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Q.6 (a) Describe digital voltmeters (DVM) and digital ammeters (DAM) with block diagrams. [7]

(b) Describe the working principle of Potential Transformers (PTs) and their applications. [7]

Q.7 (a) Explain the block diagram and working of Digital Storage Oscilloscope (DSO). [7]

(b) Explain errors and limitations in RLC measurement instruments. [7]

Q.8 (a) Explain the construction and working of True RMS meters. [7]

(b) Explain the basic components of biomedical instruments with block diagrams. [7]

Q.9 (a) Explain EEG measurement techniques and waveform analysis. [7]

(b) Explain principles of CAT (CT) scan techniques and image reconstruction. [7]

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